

Final Year Project Proposal

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Title

AI-Powered Unified Mobile Marketplace with Intelligent Comparison and Review Summarization - Includes Secure Seller Integration and Multi-Source Aggregation.

Objectives

- Create a unified platform aggregating specs, prices, and deals.
- Provide AI-driven mobile comparisons and scoring.
- Enable AI-powered review summarization from YouTube.
- Establish secure seller verification and fraud prevention.
- Deliver intelligent chatbot with multi-language support.

Problems

- **Scattered Information** – Specs, new and used prices are on different platforms.
- No AI-driven comparison or recommendation.
- Lack of verified and summarized reviews.
- High fraud risks in second-hand mobile market.
- Limited opportunities for small shopkeepers to compete online.

Solution

- Integrates GSMArena, Daraz, PriceOye, and OLX data for unified comparison.
- AI provides smart comparisons and personalized recommendations.
- Gemini API used for AI reasoning, with Hugging Face models for NLP tasks.
- Ensures seller verification and fraud prevention (CNIC-based).
- Uses encrypted data storage and escrow payments for transaction security.

Tech Stack

Frontend: Next.js, TailwindCSS, Ant Design

Backend: Node.js, Next.js APIs

Database: PostgreSQL with Prisma ORM

AI Layer: Gemini API, Hugging Face Models (chatbot, NLP, summarization)

APIs: GSMArena, Daraz, PriceOye, OLX scraping

AI Models

- **AI Recommendation Engine** - Provides personalized mobile recommendations based on user preferences, budget, and usage patterns.
- **NLP-powered Chatbot (Multi-language)** - Intelligent conversational assistant supporting multiple languages for user queries and support.
- **Price Prediction Model (ARIMA/LSTM)** - Forecasts future price trends for mobile devices using time-series analysis.
- **Review Summarization** - Extracts and summarizes insights from YouTube video transcripts and written reviews.
- **Fraud Detection (Rule-based + ML)** - Identifies suspicious listings and sellers using hybrid detection approach combining rules and machine learning.

System Architecture

